

Soil: what is in the mix?

Teacher guide and worked answers

Use evidence to show that soil contains rock material and organic matter.

40 minutes. Use Lesson.html for interactive teaching or Lesson.pptx for editable slides. Slides.pdf is the static alternative. Select one pupil response route and print the shared evidence it requires. Detailed answers below are for staff.

Scheme of work

[_passsb/inputs/Build SOW 2026-2027.xlsx](#) | BUILD Weekly - Spring | C30

Recognise that soils are made from rocks and organic matter.

Direct match to C 30. Extends the required rock/organic distinction with air, water and evidence limits; no percentage composition is assumed.

Prior learning

Rocks are natural materials with different properties.

Living things grow; fallen leaves came from plants.

An observation is something we notice using a sense or a tool.

Success evidence

Identify one piece of mineral material and one piece of organic material.

Support a choice with something observed or stated on an evidence card.

Explain that soil is a mixture; air and water can occupy spaces between solids.

Equipment

Printed evidence cards and selected response page; pencils or highlighters.

Optional: teacher-screened soil in shallow lidded transparent trays, hand lenses, wooden sticks and spoons.

Optional teacher demonstration: clear plastic cup, small soil sample and room-temperature water.

Preparation

Choose one pupil response route; all routes assess the same core science.

Print the six evidence cards and keep teacher answers separate.

For hands-on observation, screen samples for sharp objects, animal waste and heavy dust; use a known clean teaching source.

A second adult may pre-label two observable pieces. Do not depend on finding recognisable remains in every soil.

Practical care

No tasting, smelling closely or blowing soil; keep hands away from faces and wash hands afterwards.

Keep samples in trays and use tools; offer sealed-tray or paper observation to pupils avoiding touch.

Use plastic containers; an adult pours demonstration water slowly in a spill tray and wipes spills promptly.

Do not add living animals to the activity or collect unknown soil from contaminated sites.

Ready to teach alternative

Use the six printed evidence cards. Their described observations provide all information needed for every task. A bubble model substitutes for the optional demonstration; pupils label it as model evidence.

Teaching sequence

Slide	Stage	Minutes	Focus
1	Opening	0-2	Soil: what is in the mix?
2	Retrieval	2-5	Start with what we know
3	I do	5-8	Two sources, one mixture
4	I do	8-10	Zoom into soil
5	I do	10-12	A clue we cannot pick up
6	We do	12-16	The evidence desk
7	We do	16-20	Accept, reject or need more?
8	Hinge	20-23	Which account is best?
9	You do	23-25	Choose your evidence route
10	You do	Within stage	Look closely; keep a trace
11	You do	25-35	Build your explanation
12	You do	Within stage	Check your account
13	Review	35-38	Lundy loop: use your evidence
14	Exit	38-40	Three small checks

Times are a suggested 40-minute route. Slides marked Within stage share the preceding stage allocation. The optional HTML timer starts only when selected.

1 Opening Soil: what is in the mix?

Welcome pupils to a calm evidence lab. Accept initial ideas without confirming all. State the objective: use evidence to identify mineral and organic material. Choose handling or paper route privately.

2 Retrieval Start with what we know

Read each question and allow spoken, pointed or written replies. Do not treat a quiet response as no understanding. If rock/material knowledge is insecure, point to A and name a mineral grain.

Reveal: 1 Yes: a plant. 2 For example, hardness or colour. 3 See small details more clearly.

3 I do Two sources, one mixture

Point to both branches. Think aloud: this rock breaks down and changes through weathering; it supplies mineral material. A fallen leaf came from a living plant. Its remains contribute organic matter. We are tracing sources, not saying every soil has equal amounts. Our mixed classroom model is not mature soil.

4 I do Zoom into soil

Use the model, not a copied photo. Trace one mineral grain then a dead plant fragment. Point to one air pore and one water pore. State that real particles and pores differ greatly; this drawing does not give actual percentages or true sizes.

5 I do A clue we cannot pick up

Optional adult demonstration only; otherwise use this before/after model. Think aloud: I see bubbles, so there is gas. In this ordinary soil test, escaping air is a sensible explanation. The diagram is a model; it is not a record of our own experiment.

Reveal: Air was in some spaces. Water entering pushed some of it out.

6 We do The evidence desk

Pair roles: reader/observer and classifier, then swap. A: quartz identified as mineral, so mineral material. B/C: recognised plant remains, so organic. Invite a quiet pointing response plus adult-scribed reason. These are constructed examples, not observations made today.

Reveal: A: mineral. B and C: organic. Origin is stronger evidence than colour.

7 We do Accept, reject or need more?

Give each pair three positions on paper: accept, reject, need more. The wording “D must be organic” overclaims; reject the certainty or choose need more, provided reasoning says colour cannot decide. E supports escaping air; F supports water. Press for “supports” rather than “proves everything about the sample”.

Reveal: D: need more evidence. E and F: supported by the stated observations.

8 Hinge Which account is best?

Require an individual choice before discussion. If many choose A, rebuild the two-branch diagram. If C, contrast a dark mineral with the leaf-vein evidence. If secure, invite “can contain” rather than claiming every pore always contains both air and water.

A. Soil is only ground-up rock. - Mineral material is part of soil, but this misses organic material and pores.

B. Soil contains mineral and organic material; pores can contain air and water. - This includes the key solid components and the spaces between them.

C. Every dark speck is a dead plant. - Colour alone cannot tell us the material or its origin.

9 You do Choose your evidence route

Give one route only, selected with the pupil. Supported uses A-C and the word bank; standard records all six cards; stretch assesses uncertain dark grains and improves a model. Pupils may observe screened soil too, but the evidence cards are sufficient.

10 You do Look closely; keep a trace

This slide shares the 12-minute You do stage. Hands-on: use a spoon or stick to move a small amount within a tray, observe with a lens, draw one mineral-looking grain and one identifiable plant fragment if present. Do not identify every tiny piece from appearance; record uncertain where appropriate. No sample is required for paper route.

11 You do Build your explanation

Independent work time includes optional observation, route tasks and a brief check. Provide a spoken rehearsal before writing. Check that each pupil links evidence to a component, not only recites a word. Stretch pupils add air/water and a model limitation.

12 You do Check your account

Shares the independent stage. Pupils use these checks to improve one response. Accept a private spoken improvement, a moved card, or a changed label as evidence of editing.

13 Review Lundy loop: use your evidence

Hear at least two different response routes without ranking them. Use one pupil suggestion to alter the drawn model and name the change. This gives the voice an actual influence. Ask one pupil to explain why the dark grain stayed uncertain.

Reveal: Organic material; also show spaces for air and water.

14 Exit Three small checks

Collect exit page or three private spoken answers. Secure core: a mineral example and an organic example with sensible origins. If either is missing, start next lesson with A/B card matching. Air/water is extension to the core outcome, not a barrier to recording progress.

Reveal: A quartz grain; a leaf or stem fragment; air or water.

Answers and assessment

Retrieval 1-3

1 Yes, a fallen leaf came from a living plant. 2 Accept hardness, colour, texture or another suitable property; not "all rocks are heavy/hard". 3 A lens magnifies small visible details.

We do W1 / cards A-C

A mineral: quartz is identified as a mineral from rocks. B organic: recognised fallen-leaf tissue/vein. C organic: plant-stem fibres. Colour is not the deciding clue.

We do W2 / cards D-F

D remains uncertain; dark colour alone is insufficient. E supports air/gas occupying pores before water displaced it. F supports water in that damp sample. Neither E nor F tells us the percentages of each component.

Hinge

Second option is correct. Rock-only misses organic matter and pore contents. Dark colour does not show every grain is a dead plant. Reteach with the two-source diagram and cards A/B if needed.

S1-S5

S1 mineral material. S2 organic matter. S3 it came from a plant stem / has recognisable plant fibres. S4 mineral material and organic matter. S5 a space between particles containing air or water; accept a labelled sketch.

T1

A mineral - quartz origin. B organic - leaf vein/plant origin. C organic - stem/fibres. D uncertain - only colour known. E air in spaces - bubbles escape as water enters. F water - damp sample leaves water mark.

T2-T3

T2 soil also contains organic matter from living things and pores that can contain air/water. Core credit requires organic matter alongside mineral material. T3 colour is not sufficient; we need information about its material/origin or further examination. Do not demand an unavailable laboratory test name.

X1-X4

X1 include mineral grains and organic fragments, with labelled gaps. X2 air/water goes in gaps, not inside every solid particle. X3 dark minerals and organic pieces can both be dark. X4 model helps explain a mixture or pore spaces; it does not show actual sizes, proportions, organisms, chemistry or long formation processes. Accept any accurate limitation.

Exit E1-E3

E1 quartz grain, rock fragment or mineral particle. E2 leaf/stem remains or another valid example. E3 air or water. Do not demand both if the prompt is singular.

R1-R2

Reflective answers vary. Accept an evidence-related revision, e.g. "I thought soil was only rock; leaf fibres changed my mind." R2 should be a relevant question or observation; no fixed right preference.

Optional observation

Actual results vary. Record visible observations without inventing identifiable organic fragments. "Cannot tell" is scientifically valid for an unidentifiable particle; pair it with card evidence so the core objective can still be assessed.

Teaching and assessment notes

40 minutes: opening 2; retrieval 3; I do 7; We do 8; hinge 3; You do 12; review 3; exit 2. Zero-minute slides share the stage time.

The BUILD SoW uses a Year 3 science anchor presented age-appropriately for older learners. These are access routes, not diagnoses or permanent attainment groups.

Teach the word organic in its science meaning here: material from living things, not a supermarket production label. Living organisms also inhabit soil; visible leaf fragments are not all of its organic matter.

A cover teacher can teach every task with paper cards. Say: "We will use the evidence we have. If it does not decide, we will say uncertain."

Assess reasoning through a pointing choice plus spoken clue as readily as extended writing. No public ranking. Secure core is identifying mineral and organic components with a relevant example.

The three-lesson sequence moves from composition to comparative drainage to an estimated water balance. Keep the distinction between looking, testing and claiming clear.

Access and responsive teaching

Supported

Read two cards aloud at a time. Pupil points, matches or dictates one evidence reason. Use the supported response page, with mineral/organic word bank.

Standard

Classify evidence independently and write a two-part explanation that includes mineral and organic components.

Stretch

Evaluate uncertain evidence and improve a rock-only model; explain one limit of the model.

Regulation

Invite seated observation or paper evidence. Give predictable roles: observer, card mover, reporter. Reporting may be private; no timed competition or compulsory touching.

Misconceptions to check

Soil is just tiny rocks.

Mineral material is one part; organic material is also present, and spaces can contain air and water.

Check: Ask pupils to add a missing component to a rock-only model.

Dark material must be organic matter.

Colour alone does not identify what a particle is made from; dark minerals also exist.

Check: Return to the dark-grain card and ask what else we need.

Air is not part of soil because we cannot see it.

Air can occupy pores. Escaping bubbles when water enters support that explanation.

Check: Ask what occupied the spaces before the water.

Mixing leaf pieces with sand makes fully developed soil at once.

It makes a model mixture. Natural soils develop through interacting processes over time.

Check: Ask which real processes the model leaves out.

Word	Meaning
soil	A mixture of mineral and organic material, with spaces for air and water.
mineral material	Rock fragments and tiny mineral particles.
organic matter	Material from living things, including broken-down remains.
pore	A small space between soil particles.
evidence	Information that helps us decide whether an idea is supported.

Scientific references

USDA NRCS: [Urban Soil Primer, chapter 2](#)

Checked 8 September 2026. Supports mineral material, organic matter, air and water as soil components and variation between soils. Teaching diagrams and evidence cards are original simplified models, not copied figures.